SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – TECHNICAL PRESENTATION

Course Code:	ITA03	**Course Title:**	Mobile computing
CO Assessed:	CO3 – Precision (BL3)	**Assessment:**	Assessment Tool 2 – Technical Presentation
Weightage:	30 % of CIA	**Total Marks:**	20 (Answer ANY ONE)
CO3 Statement:	*Implement wireless networking and Mobile IP concepts for routing in cellular and ad hoc networks.. (BTL 3)*		
Student Name:		**Reg. No.:**	
Section / Batch:		**Date:**	

Code of Conduct

I Sybil Litzy. A (192421110) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: Sybil Litzy. A

Instructions

- Duration: 10 minutes

Faculty In-charge	Course Coordinator	Head of Department
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

Question:

1. A GSM carrier of 200 kHz is reused with cluster size N = 4 across a total 20 MHz spectrum. Derive the number of carriers per cell, and then the total voice channels per cell (8 slots/carrier), showing the full calculation chain. Discuss how this result would change under a larger cluster size like N = 7, and the resulting capacity-versus-interference trade-off.

Engineer to Excel

GSM Frequency Reuse and Capacity Calculation

Topic:
Calculation of Carriers and Voice Channels per Cell using Frequency Reuse

Introduction

- GSM divides the available spectrum into multiple radio carriers.
- Each carrier occupies **200 kHz** bandwidth.
- Frequency reuse allows the same frequencies to be used in different cells.
- The cluster size (N) determines how frequencies are shared among cells.

Problem Statement

- **Efficient Spectrum Usage:** Utilize the available GSM spectrum effectively among cells.
- **Capacity Calculation:** Calculate the carriers and voice channels per cell using the given GSM parameters.
- **Capacity vs. Interference:** Analyse how changing the cluster size affects network capacity and interference.

Engineer to Excel

Calculation

Step 1: Total Number of Carriers

Given
Total Spectrum = **20 MHz**
Carrier Bandwidth = **200 kHz = 0.2 MHz**

Formula:

Total Carriers = Total Spectrum / Carrier Bandwidth

Calculation
= 20 / 0.2
= 100 carriers

Step 2: Carriers per Cell

Cluster Size
N = 4

Formula:
Carriers per Cell
= Total Carriers / N

= 100 / 4
= 25 carriers per cell

Voice Channel Calculation

Step 3: Voice Channels per Cell
Each GSM carrier contains
8 Time Slots

Formula:
Voice Channels
= Carriers per Cell × 8

= 25 × 8
= 200 Voice Channels per Cell

Final Result
- Total Carriers = **100**
- Carriers per Cell = **25**
- Voice Channels per Cell = **200**

Calculation for Cluster Size N = 7

New Cluster Size
N = 7
Step 1:
Total Carriers = **100**
Step 2
Carriers per Cell
= 100 / 7
≈ 14 carriers
Step 3
Voice Channels
= 14 × 8
= 112 Voice Channels per Cell
Result
- Carriers per Cell ≈ **14**
- Voice Channels ≈ **112**

Capacity vs Interference Trade-off

Cluster Size	Carriers per Cell	Voice Channels	Interference	Capacity
N = 4	25	200	Higher	Higher
N = 7	14	112	Lower	Lower

Explanation

N = 4

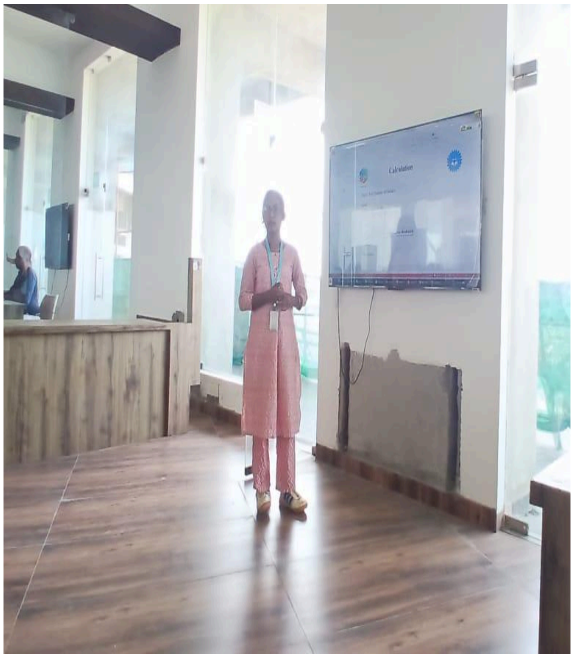
- More carriers available.
- Higher voice capacity.
- More co-channel interference.

N = 7

- Fewer carriers available.
- Lower voice capacity.
- Better signal quality.
- Less interference.

Conclusion

- A total GSM spectrum of 20 MHz provides 100 carriers, with each carrier occupying 200 kHz of bandwidth.
- For a cluster size of N = 4, each cell receives 25 carriers, resulting in 200 voice channels (25 × 8).
- For a cluster size of N = 7, each cell receives approximately 14 carriers, resulting in 112 voice channels, which reduces network capacity.
- A smaller cluster size (N = 4) increases the number of users that can be served but also increases the possibility of co-channel interference.
- A larger cluster size (N = 7) reduces interference and improves communication quality, but at the cost of lower system capacity. Therefore, the choice of cluster size should balance capacity and signal quality.


Calculation